

Logistic Regression → Classification Technique

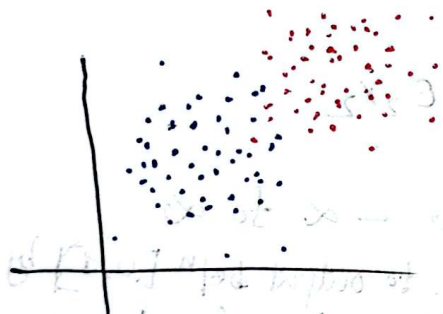
Logistic Regression

↳ Classification algorithm based upon supervised Learning.

↳ The goal of the algorithm is to do binary classification, classify the input into one of the two classes.

But this classification can be performed with any no. of categories.

↳ The model outputs the probability of a categorical target variable Y belonging to a certain class.



• Class A

• Class B

Example

Email - spam or not ?

Is this fruit ripe or not ?

Is it going to rain or not ?

Ex:

Fruits Dataset

x_1 (weight)	x_2 (sweetness)	Label
8	7	1
10.1	9	1
12	11	1
8.5	3.8	1
3.5	5	0
2.5	2	0
6.4	4	0
9.7	8	1

1 = Apple

0 = Orange

$[x_1, x_2, \dots, x_n]$
input with
n features

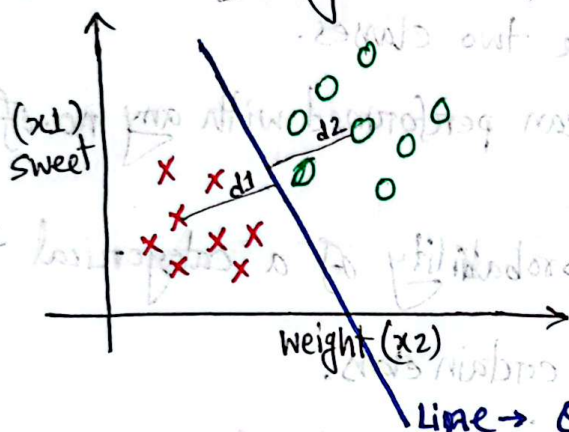
Model
(Hypothesis)

→ $y \in [0, 1]$
probability
value betⁿ
0 to 1.

Step-01

Hypothesis function

Logistic regression is also a parametric method, which means it makes an assumption about the form of the function relating X and Y .



$$d1 = \frac{\theta_0 + \theta_1 x_1 + \theta_2 x_2}{\sqrt{\theta_1^2 + \theta_2^2}}$$
$$d2 = \frac{\theta_0 + \theta_1 x'_1 + \theta_2 x'_2}{\sqrt{\theta_1^2 + \theta_2^2}}$$

Same

So, $d \propto \theta_0 + \theta_1 x_1 + \theta_2 x_2$

So, $d = \theta_0 + \theta_1 x_1 + \theta_2 x_2$

$\Rightarrow d = \theta^T X \rightarrow -\infty \text{ to } \infty$

Logit Model $\rightarrow$ Makes sure to output betⁿ $[0, 1]$ by applying sigmoid.

Sigmoid $\rightarrow$ Squashes the ~~real~~ real values in the range $[0, 1]$.

$$s(z) = \frac{1}{1 + e^{-z}}$$

So, $y_p = h_\theta(x) = \sigma(\theta^T x)$

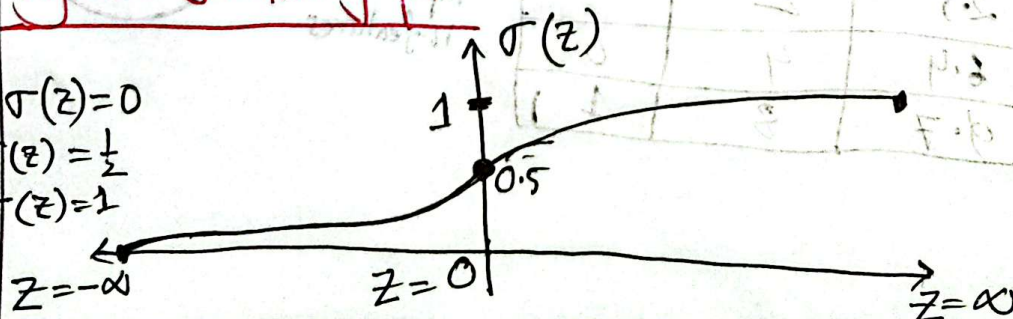
$\Rightarrow h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}} = \frac{1}{1 + e^{-\sum_{i=0}^n \theta_i x_i}}$

Sigmoid function graph

If $z = -\infty$, $\sigma(z) = 0$

$z = 0$, $\sigma(z) = \frac{1}{2}$

$z = \infty$, $\sigma(z) = 1$



Step-02 Loss function (Binary Crossentropy)

~~Loss~~ ^{or} Log Loss

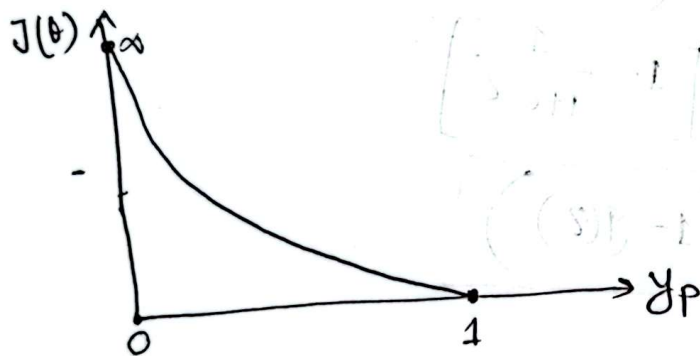
$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m \left(y^{(i)} \log y_p^{(i)} + (1-y^{(i)}) \cdot \log(1-y_p^{(i)}) \right)$$

Case-I: $y^{(i)} = [1, 1, 1, 1, 1]$

$$\therefore J(\theta) = -\frac{1}{m} \sum_{i=1}^m y^{(i)} \log y_p^{(i)}$$

if, $y_p = 1$, ~~Loss~~ $J(\theta) = 0$

if $y_p = 0$, $\text{Loss}, J(\theta) = +\infty$

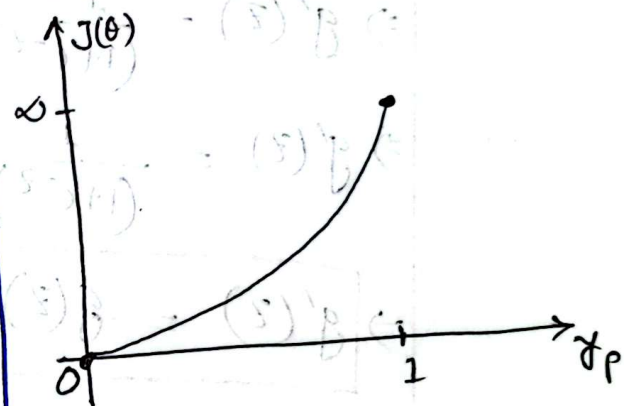


Case-II: $y^{(i)} = [0, 0, 0, 0, 0]$

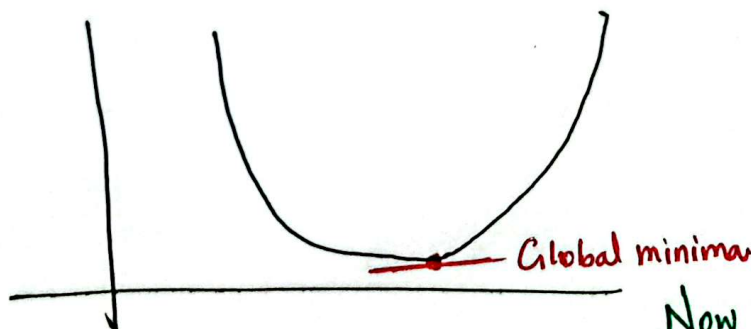
$$\therefore J(\theta) = -\frac{1}{m} \sum_{i=1}^m (1-y^{(i)}) \log(1-y_p^{(i)})$$

if $y_p = 0$, $\text{Loss}, J(\theta) = 0$

if $y_p = 1$, $\text{Loss}, J(\theta) = +\infty$



Combine



Now we can apply
Gradient Descent

Step-03 Training (Gradient descent)

$$\theta_j = \theta_j - \eta \cdot \frac{\partial J(\theta)}{\partial \theta_j} \quad \text{where } J(\theta) \text{ is binary Crossentropy}$$

Derivative of a sigmoid function

We know

$$\frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2}$$

$$g(z) = \frac{1}{1+e^{-z}}$$

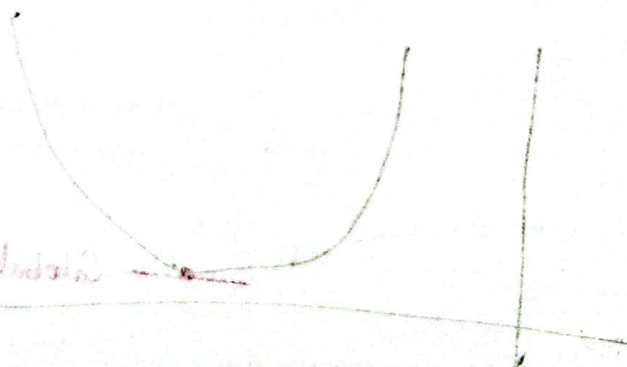
$$\Rightarrow g'(z) = -\frac{1}{(1+e^{-z})^2} \cdot e^{-z} \cdot (-1)$$

$$\Rightarrow g'(z) = \frac{e^{-z}}{(1+e^{-z})^2}$$

$$\Rightarrow g'(z) = \frac{1}{(1+e^{-z})} \cdot \frac{e^{-z}}{(1+e^{-z})}$$

$$\Rightarrow g'(z) = \frac{1}{(1+e^{-z})} \cdot \left[1 - \frac{1}{1+e^{-z}} \right]$$

$$\Rightarrow g'(z) = g(z) \cdot (1 - g(z))$$



Now,

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \cdot \log y_p^{(i)} + (1-y^{(i)}) \cdot \log (1-y_p^{(i)}) \right]$$

~~$$J(\theta)$$~~

$$\Rightarrow \frac{\partial J(\theta)}{\partial \theta_j} = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \cdot \frac{1}{y_p^{(i)}} \cdot \frac{\partial y_p^{(i)}}{\partial \theta_j} + (1-y^{(i)}) \cdot \frac{1}{1-y_p^{(i)}} \cdot \left(-\frac{\partial y_p^{(i)}}{\partial \theta_j} \right) \right]$$

$$\Rightarrow \frac{\partial J(\theta)}{\partial \theta_j} = -\frac{1}{m} \sum_{i=1}^m \left[\frac{y^{(i)}}{y_p^{(i)}} - \frac{1-y^{(i)}}{1-y_p^{(i)}} \right] \cdot \frac{\partial y_p^{(i)}}{\partial \theta_j}$$

we know,
 $y_p^{(i)} = h_\theta(x^{(i)})$
 $= \sigma(\theta^T x^{(i)})$

Here, $\frac{\partial y_p^{(i)}}{\partial \theta_j} = \sigma'(\theta^T x^{(i)})$

$$= \sigma(\theta^T x^{(i)}) \cdot (1 - \sigma(\theta^T x^{(i)})) \cdot x_j^i$$

~~$$= y_p^{(i)} \cdot (1 - y_p^{(i)}) x_j^i$$~~

$$\text{So, } \frac{\partial J(\theta)}{\partial \theta_j} = -\frac{1}{m} \sum_{i=1}^m \left[\frac{y^{(i)} - y^{(i)} y_p^{(i)} - y_p^{(i)} + y^{(i)} y_p^{(i)}}{y_p^{(i)} \cdot (1 - y_p^{(i)})} \right] \cdot y_p^{(i)} \cdot (1 - y_p^{(i)}) \cdot x_j^i$$

$$\Rightarrow \frac{\partial J(\theta)}{\partial \theta_j} = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} - y_p^{(i)}] x_j^i$$

$$\Rightarrow \frac{\partial J(\theta)}{\partial \theta_j} = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} - h_\theta(x^{(i)})] \cdot x_j^i$$

~~KA~~
~~2~~